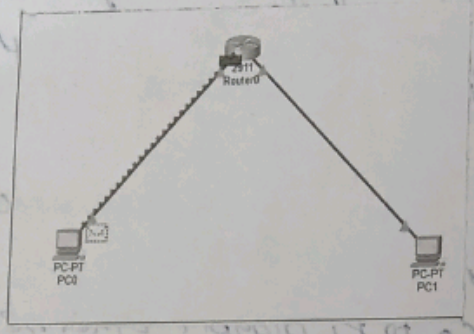
EXP NO: 10 a) INTERNETWORKING WITH ROUTERS IN CISCO PACKET TRACER

Aim:
To design and configure a simple networking using a router and establish communication between two different networks

PROCEDURE:

1. Open cisco packet tracer and place one router and two PCs in the work space.
2. Connect PC0 to R1 Gigabit Ethernet 0/0 and PC1 to R1 gigabit Ethernet 0/1 using copper-straight through cables.
3. configure R1 G0/0 with:
   IP: 192.168.10.1
   Subnet mask: 255.255.255.0
4. configure R1 G0/1 with:
   IP: 192.168.20.1
   Subnet mask: 255.255.255.0
5. Enable both router interfaces using no shut down command.
6. configure PC0 with:
   IP: 192.168.10.2
   Subnet mask: 255.255.255.0
   Gateway: 172.168.10.1
7. Configure PC1 with:
   IP: 192.168.20.2
   Subnet mask: 255.255.255.0
   Gateway: 192.168.20.1
8. Test connectivity by pinging from PC0 to PC1 using: ping 192.168.20.2
9. switch to simulation mode & send a simple PDU from PC0 to PC1




```
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface              IP-Address      OK? Method Status                Protoco
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#


Router con0 is now available
```

Result:

Thus a simple Internetwork was successfully configured using a router, and communication between PC0 and PC1 was successfully verified

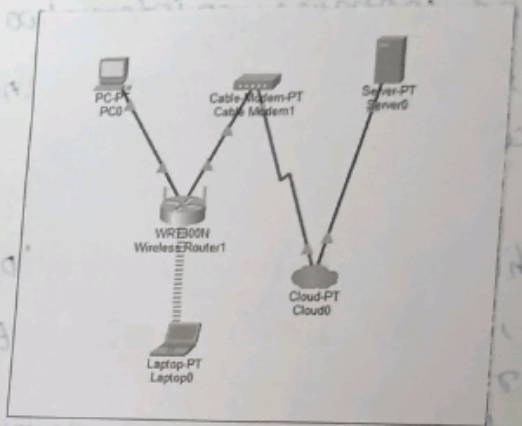
b) DESIGN AND CONFIGURE AN INTERNETWORK USING WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

Aim: To design and configure an internetwork using wireless router, DHCP server and internet cloud in cisco packet tracer.

PROCEDURE:

1. Create the topography using a PC, laptop, wireless router, cable modem, internet cloud & cisco.com server.
2. Connect the devices using appropriate cable
3. Configure the laptop ~~to connect~~ to wireless router using the SSID HomeNetwork and enable DHCP.
4. Configure the laptop to connect to the wireless network
5. Configure PC0 to obtain its IP address automatically using DHCP
6. Configure the Internet cloud and cable modem connections.
7. Configure the cisco.com server as a DHCP and DNS server.
8. Assign the server IP address 208.67.220.220
9. Verify the IP configuration of PC0 using ip config /all.
10. Test connectivity using ping 208.67.220.22 and ping cisco.com.

TOPOLOGY:




```
C:\>
C:\>ping 208.67.220.220

Pinging 208.67.220.220 with 32 bytes of data:

Request timed out.
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping cisco.com

Pinging 208.67.220.220 with 32 bytes of data:

Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Result:

Thus an internetwork using a wireless router DHC Server and internet cloud was successfully configured and connectivity to the cisco.com server was verified.

C) DESIGNING AN INTER-NETWORK USING SWITCH, ROUTER & ETHERNET

Aim:

To design and configure an internetwork using a switch, router, and ethernet cables in cisco packet tracer.

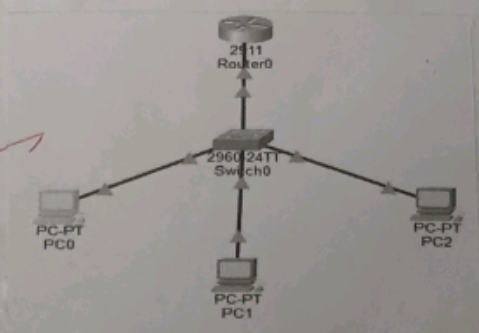
THEORY:

A switch connects multiple devices within a LAN while a router connects and manages communication between networks. Ethernet cables are used to establish wired connections between the devices.

PROCEDURE:

1. place 1 router, 1 switch and 3 PCs.
2. connet them using copper straight through cables.
3. Configure router G0/0 as 192.168.30.1/24.
4. Assign PC0, PC1 and PC2 the IPs 192.168.30.2, .3 and .4.
5. set the default gatway as 192.168.30.1
6. use ping to verify connectivity between the PCs.

TOPOLOGY:



```
C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=26ms TTL=125
Reply from 20.0.0.2: bytes=32 time=18ms TTL=125
Reply from 20.0.0.2: bytes=32 time=14ms TTL=125
Reply from 20.0.0.2: bytes=32 time=14ms TTL=125

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 26ms, Average = 18ms

C:\>tracert 20.0.0.2

Tracing route to 20.0.0.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.0.0.1
  2    1 ms      0 ms      1 ms      192.168.1.250
  3   10 ms      1 ms      1 ms      192.168.1.245
  4    5 ms      0 ms      3 ms      20.0.0.2

Trace complete.
```

```
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

Result:

Thus the internetwork was successfully conf
using a switch, router and ethernet cables, and
connectivity was verified using the ping
command.